IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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https://github.com/ddlbnt/IBM-Applied-Data-Science-Capstone-Project-ddlbnt.git

Project Overview

- Data Collection – Gather data from SpaceX API and Wikipedia web scraping.
- Data Wrangling – Clean data, handle missing values, create Class column (0/1).
- EDA – Analyze data using Seaborn, Matplotlib, and SQL queries.
- Visual Analytics – Interactive Folium maps and Dash dashboard.
- ML Prediction – Predict launch success with 4 ML models; evaluate using Accuracy/F1, Confusion Matrix, ROC curve.

Executive Summary

- SpaceX API – Collect launch data.
- Web Scraping – Gather additional data from Wikipedia.
- Data Wrangling – Clean data, handle missing values, create Class column.
- EDA – Analyze data with charts and SQL queries.
- Visual Analytics – Interactive Folium maps and Dash dashboard.
- ML Prediction – Predict launch success using 4 ML models.

Introduction

- Background – Analyze SpaceX Falcon 9 launches to understand patterns of success and failure.
- Problem Statement – Predict launch success and provide actionable insights.
- Objective – Explore data, identify trends, and apply predictive models for decision-making.

Data Collection– SpaceX API

- Purpose– Collect launch data from SpaceX API.
- Steps:
  - Send GET requests to SpaceX API endpoints.
  - Process JSON responses and convert to Pandas DataFrame.
  - Inspect data with head() to verify results.
- Reference File: SpaceX_Data_Collection_API.ipynb
- GitHub URL: https://github.com/ddtbnt/IBM-Applied-Data-Science-Capstone-Project-ddtbnt/blob/main/SpaceX-API/SpaceX_Data_Collection_API.ipynb
- Tip: Include a screenshot of API call code and sample DataFrame output.

Data Collection– Web Scraping

- Purpose– Collect additional Falcon 9 launch data from Wikipedia.
- Steps:
  - Use BeautifulSoup to scrape tables from Wikipedia pages.
  - Process HTML tables into clean Pandas DataFrame.
  - Verify and inspect data for completeness.
- Reference File: SpaceX_Web_Scraping.ipynb
- GitHub URL: https://github.com/ddtbnt/IBM-Applied-Data-Science-Capstone-Project-ddtbnt/blob/main/Web-Scraping/SpaceX_Web_Scraping.ipynb
- Tip: Include screenshot of scraping code and sample scraped table.

Data Wrangling Methodology

- Purpose – Clean and prepare the collected data for analysis.
- Steps:
  - Handle missing values and remove duplicates.
  - Create Class column: 0 =failed, 1=success.
  - Merge SpaceX API data with Web Scraped data into a unified DataFrame.
- Reference File: SpaceX_Data_Wrangling.ipynb
- GitHub URL: https://github.com/ddtbnt/IBM-Applied-Data-Science-Capstone-Project-ddtbnt/blob/main/Data-Wrangling/SpaceX_Data_Wrangling.ipynb
- Tip: Include screenshot of key cleaning code and example table after wrangling.
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EDA with Data Visualization

- Purpose – Explore patterns and trends in the data using visualizations.
- Steps / Visuals:
  - Distribution of payload and success rate.
  - Catplot, scatterplot, and bar charts using Seaborn. Extract insights from visual trends.
- Reference File: SpaceX_EDA_Visualization.ipynb
- GitHub URL: https://github.com/ddtbnt/IBM-Applied-Data-Science-Capstone-Project- ddtbnt/blob/main/EDA-with-Visualization/SpaceX_EDA_Visualization.ipynb
- Tip: Include 2–3 key charts as screenshots and brief notes on insights.
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EDA with SQL

- Purpose – Analyze SpaceX data using SQL queries to extract insights.
- Steps / Queries:
- Use SELECT, SUM, COUNT, GROUP BY to summarize launches.
- Identify launches per year, success rates per launch site.
- Explore trends in payloads and outcomes.
- Reference File: SpaceX_EDA_SQL.ipynb
- GitHub URL: https://github.com/ddtbnt/IBM-Applied-Data-Science-Capstone-Project-ddtbnt/blob/main/EDA%20with%20SQL/SpaceX_EDA_SQL.ipynb
- Tip: Include screenshots of SQL code and result tables.

Interactive Visual Analytics

- Purpose – Visualize SpaceX launch data interactively.
- Steps / Tools:
  - Folium maps: Marker clusters, distance lines between launch sites.
  - Dash dashboard: Filter by payload, launch site, and booster version.
  - Explore patterns and correlations interactively.
- Reference Files: SpaceX_Folium_Maps.ipynb & SpaceX_Dash_App.py
- GitHub URLs:
  - Folium: SpaceX_Folium_Maps.ipynb
  - Dash: SpaceX_Dash_App.py
- Tip: Include screenshots of interactive map and Dash dashboard.

EDA Visualization Results

- Purpose – Show insights from data visualizations.
- Key Visuals:
  - Scatter plots: Payload vs launch outcome.
  - Bar charts: Success rate by booster version.
  - Catplots: Launch site success comparison.
- Reference File: SpaceX_EDA_Visualization.ipynb
- GitHub URL: https://github.com/ddtbnt/IBM-Applied-Data-Science-Capstone-Project-ddtbnt/blob/main/EDA-with-Visualization/SpaceX_EDA_Visualization.ipynb
- Tip: Include screenshots of 2–3 charts with brief insight notes.

EDA SQL Results

- Purpose – Summarize insights using SQL queries.
- Key Queries / Insights:
  - Total launches per year and per launch site.
  - Success vs failure rates by site.
  - Average payloads and other summary statistics.
- Reference File: SpaceX_EDA_SQL.ipynb
- GitHub URL: https://github.com/ddtbnt/IBM-Applied-Data-Science-Capstone-Project-ddtbnt/blob/main/EDA%20with%20SQL/SpaceX_EDA_SQL.ipynb
- Tip: Include screenshots of SQL queries and resulting summary tables.

Folium Map Results

- Purpose – Display interactive map insights for SpaceX launches.
- Key Visuals / Features:
  - Marker clusters for launch sites.
  - Lines showing distances between sites.
  - Proximity analysis of launch locations.
- Reference File: SpaceX_Folium_Maps.ipynb
- GitHub URL: https://github.com/ddtbnt/IBM-Applied-Data-Science-Capstone-Project-ddtbnt/blob/main/Folium/SpaceX_Folium_Maps.ipynb
- Tip: Include screenshots of interactive map with clusters and lines.

Plotly Dash Results

Purpose – Showcase interactive dashboard for launch data analysis.

- Key Features / Visuals:
- Pie chart: Success vs failure launches.
- Scatter chart: Payload vs launch outcome.
- Interactive filters: Filter by payload, site, booster version.
- Reference File: SpaceX_Dash_App.py
- GitHub URL : https://github.com/ddtbnt/IBM-Applied-Data-Science-Capstone-Project- ddtbnt/blob/main/EDA-with-Visualization/SpaceX_Dash_App.py
- Tip: Include screenshot of Dash dashboard showing pie and scatter charts with filters.

Predictive Analysis Results

- Purpose – Predict Falcon 9 launch success using Machine Learning models.
- Models Used: Logistic Regression, SVM, Decision Tree, Random Forest.
- Evaluation Metrics:
- Confusion Matrix
- Accuracy / F1 Score
- ROC Curve
- Highlight: Best performing model and key insights.
- Reference File: SpaceX_Machine_Learning_Part_5.ipynb
- GitHub URL: https://github.com/ddtbnt/IBM-Applied-Data-Science-Capstone-Project-ddtbnt/blob/main/SpaceX_Machine_Learning_Part_5.ipynb
- Tip: Include screenshot of confusion matrix, accuracy table, and ROC curve.